



2807 West County Road 75
Monticello, MN 55362

April 25, 2024

L-MT-24-012
10 CFR 50.73

ATTN: Document Control Desk
U.S. Nuclear Regulatory Commission
Washington, DC 20555-0001

Monticello Nuclear Generating Plant
Docket No. 50-263
Renewed Facility Operating License No. DPR-22

Monticello Nuclear Generating Plant Licensee Event Report 2024-001-00

Northern States Power Company, a Minnesota corporation, doing business as Xcel Energy (hereafter "NSPM"), hereby submits Licensee Event Report (LER) 50-263/2024-001-00 per 10 CFR 50.73(a)(2)(i)(B) and 10 CFR 50.73(a)(2)(iv)(A).

If you have any questions about this submittal, please contact Carrie Seipp, Senior Regulatory Engineer, at 612-330-5576.

Summary of Commitments

This letter makes no new commitments and no revisions to existing commitments.

A handwritten signature in black ink, appearing to read 'Greg D Brown'.

Greg D Brown
Plant Manager, Monticello Nuclear Generating Plant
Northern States Power Company – Minnesota

Enclosure

cc: Administrator, Region III, USNRC
Project Manager, Monticello, USNRC
Resident Inspector, Monticello, USNRC
State of Minnesota

ENCLOSURE

**MONTICELLO NUCLEAR GENERATING PLANT
LICENSEE EVENT REPORT 50-263/2024-001-00**

3 pages follow

NRC FORM 366 (04-02-2024)		U.S. NUCLEAR REGULATORY COMMISSION		APPROVED BY OMB: NO. 3150-0104		EXPIRES: 04/30/2027					
		LICENSEE EVENT REPORT (LER) (See Page 2 for required number of digits/characters for each block) (See NUREG-1022, R.3 for instruction and guidance for completing this form http://www.nrc.gov/reading-rm/doc-collections/nuregs/staff/sr1022/r3/)									
1. Facility Name Monticello Nuclear Generating Plant				☒ 050 ☐ 052		2. Docket Number 263		3. Page 1 OF 3			
4. Title Reactor Scram, Containment Isolation, and Cooldown Rate outside of limits following technician adjustment of wrong component											
5. Event Date			6. LER Number			7. Report Date			8. Other Facilities Involved		
Month	Day	Year	Year	Sequential Number	Revision No.	Month	Day	Year	Facility Name	☐ 050 Docket Number	
02	28	24	24	- 001 -	00	04	25	24	Facility Name	☐ 052 Docket Number	
9. Operating Mode 1						10. Power Level 100					
11. This Report is Submitted Pursuant to the Requirements of 10 CFR §: (Check all that apply)											
10 CFR Part 20		☐ 20.2203(a)(2)(vi)		10 CFR Part 50		☐ 50.73(a)(2)(ii)(A)		☐ 50.73(a)(2)(viii)(A)		☐ 73.1200(a)	
☐ 20.2201(b)		☐ 20.2203(a)(3)(i)		☐ 50.36(c)(1)(i)(A)		☐ 50.73(a)(2)(ii)(B)		☐ 50.73(a)(2)(viii)(B)		☐ 73.1200(b)	
☐ 20.2201(d)		☐ 20.2203(a)(3)(ii)		☐ 50.36(c)(1)(ii)(A)		☐ 50.73(a)(2)(iii)		☐ 50.73(a)(2)(ix)(A)		☐ 73.1200(c)	
☐ 20.2203(a)(1)		☐ 20.2203(a)(4)		☐ 50.36(c)(2)		☒ 50.73(a)(2)(iv)(A)		☐ 50.73(a)(2)(x)		☐ 73.1200(d)	
☐ 20.2203(a)(2)(i)		10 CFR Part 21		☐ 50.46(a)(3)(ii)		☐ 50.73(a)(2)(v)(A)		10 CFR Part 73		☐ 73.1200(e)	
☐ 20.2203(a)(2)(ii)		☐ 21.2(c)		☐ 50.69(g)		☐ 50.73(a)(2)(v)(B)		☐ 73.77(a)(1)		☐ 73.1200(f)	
☐ 20.2203(a)(2)(iii)				☐ 50.73(a)(2)(i)(A)		☐ 50.73(a)(2)(v)(C)		☐ 73.77(a)(2)(i)		☐ 73.1200(g)	
☐ 20.2203(a)(2)(iv)				☒ 50.73(a)(2)(i)(B)		☐ 50.73(a)(2)(v)(D)		☐ 73.77(a)(2)(ii)		☐ 73.1200(h)	
☐ 20.2203(a)(2)(v)				☐ 50.73(a)(2)(i)(C)		☐ 50.73(a)(2)(vii)					
☐ OTHER (Specify here, in abstract, or NRC 366A).											
12. Licensee Contact for this LER											
Licensee Contact Carrie Seipp, Senior Nuclear Regulatory Engineer									Phone Number (Include area code) 612-330-5576		
13. Complete One Line for each Component Failure Described in this Report											
Cause	System	Component	Manufacturer	Reportable to IRIS		Cause	System	Component	Manufacturer	Reportable to IRIS	
14. Supplemental Report Expected						15. Expected Submission Date			Month	Day	Year
☒ No		☐ Yes (If yes, complete 15. Expected Submission Date)									
16. Abstract (Limit to 1326 spaces, i.e., approximately 13 single-spaced typewritten lines) On February 28, 2024 at 0839 CST, with Monticello Nuclear Generating Plant in Mode 1 at 100 percent power, an inadvertent Anticipated Transient Without Scram (ATWS) system initiation resulted in trip of both Reactor Recirculation pumps, venting of the Scram Air Header, and Control Rod insertion. Control Room Operators inserted a manual Reactor Scram. Containment Isolation valves actuated and closed on a valid Group 2 signal post Reactor Scram. On March 08, 2024 at 1400 CST it was identified that within an hour following the Reactor Scram, the bottom head of the Reactor Pressure Vessel exceeded the Pressure and Temperature Limit cooldown rate of less than or equal to 100 degrees Fahrenheit per hour. The event was caused when a technician did not perform verification that the physical component matched the Surveillance Procedure step. The technician adjusted the potentiometer in the C module rather than the A module while the A channel was in trip which made up the required two trip signals and actuated the ATWS system. The Surveillance Procedure will be updated in a planned procedure change to improve human performance factors and to remove the steps to adjust the potentiometers in the A and C modules. An analytical evaluation determining the effects of the out-of-limit condition on the structural integrity of the Reactor Coolant System will be submitted to the Nuclear Regulatory Commission for review and approval per 10 CFR 50.55a(b)(2)(xliii).											

**LICENSEE EVENT REPORT (LER)
CONTINUATION SHEET**

(See NUREG-1022, R.3 for instruction and guidance for completing this form
<http://www.nrc.gov/reading-rm/doc-collections/nuregs/staff/sr1022/r3/>)

Estimated burden per response to comply with this mandatory collection request: 80 hours. Reported lessons learned are incorporated into the licensing process and fed back to industry. Send comments regarding burden estimate to the FOIA, Library, and Information Collections Branch (T-6 A10M), U. S. Nuclear Regulatory Commission, Washington, DC 20555-0001, or by email to Infocollects.Resource@nrc.gov, and the OMB reviewer at: OMB Office of Information and Regulatory Affairs, (3150-0104), Attn: Desk Officer for the Nuclear Regulatory Commission, 725 17th Street NW, Washington, DC 20503. The NRC may not conduct or sponsor, and a person is not required to respond to, a collection of information unless the document requesting or requiring the collection displays a currently valid OMB control number.

1. FACILITY NAME		☒ 050	2. DOCKET NUMBER	3. LER NUMBER		
Monticello Nuclear Generating Plant		☐ 052	263	YEAR	SEQUENTIAL NUMBER	REV NO.
				24	001	00

NARRATIVE**EVENT DESCRIPTION**

On February 28, 2024 at 0839 CST, with Monticello Nuclear Generating Plant (MNGP) in Mode 1 at 100 percent power, an inadvertent Anticipated Transient Without Scram (ATWS) [EIS CODE: JC] system initiation resulted in trip of both Reactor Recirculation [EIS CODE: AD] pumps, venting of the Scram Air Header, and Control Rod insertion. Control Room Operators inserted a manual Reactor Scram. Containment Isolation [EIS CODE: JM] valves actuated and closed on a valid Group 2 signal post Reactor Scram. On March 08, 2024 at 1400 CST it was identified that within an hour following the Reactor Scram, the bottom head of the Reactor Pressure Vessel [EIS CODE: SB] exceeded the Pressure and Temperature Limit cooldown rate of less than or equal to 100 degrees Fahrenheit per hour.

This event is reportable under 10 CFR 50.73(a)(2)(iv)(A) for an event or condition that resulted in the manual actuation of the Reactor Protection System (RPS) [EIS CODE: JC] including Reactor Scram and automatic actuation of Containment Isolation signals affecting Containment Isolation valves in more than one system.

This event is reportable under 10 CFR 50.73(a)(2)(i)(B) for an operation or condition which was prohibited by Technical Specification Limiting Condition of Operation (LCO) 3.4.9, Reactor Coolant System (RCS) Pressure and Temperature (P/T) Limits, specifically the requirement to determine the RCS is acceptable for continued operation within 72 hours of not meeting the Pressure and Temperature limit cooldown rate per required action A.2.

EVENT ANALYSIS

The function of the ATWS system is to mitigate the consequences of a failure of the RPS to achieve reactor shutdown when required. The ATWS system consists of two separately powered trip systems, Division I and Division II, each made up of two sub-channels: A and C for Division I and B and D for Division II. Each sub-channel receives an input from an independent sensor monitoring each of the ATWS trip parameters. A trip occurring in both sub-channels of logic Division I or a trip occurring in both sub-channels of logic Division II will cause an ATWS trip which opens both 11 and 12 Recirculation MG set generator field breakers and causes Control Rod insertion by venting the Scram Air Header.

During performance of the quarterly Technical Specification Surveillance Procedure, ATWS - Recirc Trip for Reactor Pressure and Level Trip Unit Test and Calibration, a human performance error occurred. There were no systems, structures, or components that were inoperable at the start of the event that contributed to the event.

During the first quarter of each year during the quarterly Surveillance Requirement 3.3.4.1.6 performance of the logic system functional test on A sub-channel ATWS RCP Trip Reactor Pressure relay, the Surveillance Procedure directs the MNGP Instrumentation and Control Technician to adjust the module's trip adjusting potentiometer. This step is done in accordance with an obsolete manufacturer's recommendation in order to preclude potential equipment degradation. The Technician successfully performed the functional test on A sub-channel and then, with A sub-channel remaining in trip, obtained a different tool to perform the potentiometer adjustment. The Technician did not verify the physical component to the procedural step and incorrectly adjusted the potentiometer of the adjacent C module, which is located in the same panel as the intended A module, which cause the trip of the C sub-channel. As both A sub-channel and C sub-channel were in trip, the Division I ATWS system actuated. The inadvertent ATWS system initiation resulted in trip of both Reactor Recirculation pumps, venting of the Scram Air Header, and Control Rod insertion. Control Room Operators inserted a manual Reactor Scram. Containment Isolation valves actuated and closed on a valid Group 2 signal post Reactor Scram.

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1. FACILITY NAME Monticello Nuclear Generating Plant	☒ 050	2. DOCKET NUMBER 263	YEAR 24	3. LER NUMBER SEQUENTIAL NUMBER 001	REV NO. 00
	☐ 052		-	-	

NARRATIVE**EVENT ANALYSIS (continued)**

Within an hour following the Reactor Scram, the bottom head of the Reactor Pressure Vessel exceeded the Pressure and Temperature Limit cooldown rate of less than or equal to 100 degrees Fahrenheit per hour due to the trip of both Reactor Recirculation Pumps and the subsequent Reactor Pressure Vessel stratification. The cooldown rate throughout the beltline and upper portions of the Reactor Pressure Vessel were appropriately less than or equal to 100 degrees Fahrenheit per hour. Technical Specification 3.4.9, RCS Pressure and Temperature (P/T) Limits, requires that RCS pressure, RCS temperature, RCS heat up and cooldown rates, and the recirculation pump starting temperature requirements be maintained within the limits specified in the Pressure and Temperature Limits Report (PTLR). This LCO is applicable at all times. When in Mode 1, 2, or 3, if a limit is exceeded, required action A.1 to restore parameters within limits is required to be completed within 30 minutes and required action A.2 to determine whether the RCS is acceptable for continued operation is required to be completed within 72 hours. Parameters were not restored within 30 minutes and due to the Reactor Scram, the Unit was placed in the appropriate Modes. It was not known that the cool down limit had been exceeded when the Unit entered Mode 2 on March 2, 2024 at 1610 CST and Mode 1 on March 3, 2024 at 0234 CST. This condition was identified on March 8, 2024 at 1400 CST and Condition A.2 was entered. The determination that the RCS was acceptable for continued operation was completed on March 8, 2024 at 2200 CST and Condition A.2 was exited.

ASSESSMENT OF SAFETY CONSEQUENCES

All safety systems functioned properly during the transient. All control rods fully inserted during the Reactor Scram. There were no radiological or industrial impacts associated with this event. The health and safety of the public and site personnel were not impacted during this event.

An evaluation determined that sufficient margin exists for the bottom of the Reactor Pressure Vessel between previously analyzed conditions in the PTLR and the cooldown rate experienced on February 28, 2024.

CAUSE OF THE EVENT

The event was caused when a technician did not perform verification that the physical component matched the Surveillance Procedure step. The technician adjusted the potentiometer in the C module rather than the A module while the A channel was in trip which made up the required two trip signals and actuated the ATWS system.

CORRECTIVE ACTIONS

The Surveillance Procedure will be updated in a planned procedure change to improve human performance factors and to remove the steps to adjust the potentiometers in the A and C modules.

An analytical evaluation determining the effects of the out-of-limit condition on the structural integrity of the RCS, as described in American Society of Mechanical Engineers Boiler and Pressure Vessel Code, Section XI article IWB-3720(a) will be submitted to the NRC for review and approval per 10 CFR 50.55a(b)(2)(xlili).

PREVIOUS SIMILAR EVENTS

There were no previous similar events in the past three years.